Semester III													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24INT152T	Rural Internship	0	0	0	0	1				50	50	100
24MA201T	Mathematics-III	3	1	0	4	4	25	25	50	--	--	100
24PE201T	Introduction to Artificial Intelligence	3	0	0	3	3	25	25	50	--	--	100
25PE201T	Fundamentals of Fluid Mechanics for Petroleum Engineers	3	0	0	3	3	25	25	50	--	--	100
24PE203T	Thermodynamics of Petroleum Reservoir Fluids	2	1	0	3	3	25	25	50	--	--	100
24PE204T	Geomechanics and Strength of Materials	2	1	0	3	3	25	25	50	--	--	100
24PE204P	Geomechanics and Strength of Materials Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE205T	Petroleum Exploration	3	0	0	3	3	25	25	50	--	--	100
24PE205P	Petroleum Exploration Laboratory	0	0	2	2	1	--	--	--	50	50	100
	Total	**16**	**3**	**4**	**23**	**22**						

Semester IV													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
	Open Elective 1 (From Other School)	3	0	0	3	3	25	25	50	--	--	100
24PE206T	Industry 4.0	2	0	0	2	2	25	25	50	--	--	100
24PE206P	Industrial 4.0 Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE207T	Drilling Engineering	3	1	0	4	4	25	25	50	--	--	100
24PE207P	Drilling Engineering Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE208T	Reservoir Engineering	3	1	0	4	4	25	25	50	--	--	100
24PE209T	Well Log and Formation Evaluation	3	1	0	4	4	25	25	50	--	--	100
24PE202T	Heat and Mass Transfer	2	1	0	3	3	25	25	50	--	--	100
24PE210T	Earth Science and Hydrocarbon Exploration Field Work	0	0	6	0	3	--	--	--	50	50	100
	Total	**16**	**4**	**10**	**24**	**25**						

24INT152					Rural Internship					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	0	1	0	-	-	-	50	50	100

COURSE OBJECTIVES

- The Internship aims at exploring the students to the life, culture and issues of rural India.
- The Internship exposes students to the socio-economic aspects of Rural India

RURAL INTERNSHIP

During the internship, students will be associated with rural India, exploring

a) Rural Society
b) Rural Life
c) Rural Culture
d) Rural Development Issues
e) Rural Economy

Additionally, it also develops inter-personal skills and allied aspect.

Max <2 Weeks>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 -Summarize Rural Life of India for creating sense of belongingness for the community
CO2 -Demonstrate leadership qualities and societal responsibility
CO3- Appraise concepts of rural life and rural society for sustainable development
CO4 –Evaluate skills on participatory methodologies and tools used in rural development
CO5 -.Discover cross-cultural learning on rural and development issues for inter-personal growth
CO6 -Enable the students to appreciate the importance of agriculture, artisans and rural entrepreneurs in Rural India

24MA201T					Mathematics – III					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

COURSE OBJECTIVES

- The primary objective of this course is to provide students with a comprehensive understanding of advanced mathematical transforms and their applications in engineering.
- To use the concept of Laplace transform and its significance in solving differential equations.
- Understand the fundamental concepts of difference equation and Z-transform.
- To use this basic course in upcoming courses in respective specializations in higher classes.

Unit I: Laplace Transform **Hours: 14**

Piecewise continuous functions and exponential functions, Definition, Existence and Properties of Laplace transforms, Heaviside function, Impulse function, Inverse Laplace transform, Properties of inverse Laplace transforms, Convolution theorem, Applications of Laplace Transforms in solving differential equations.

Unit II: Difference Equation and Z-Transform **Hours: 14**

First and second order difference equation with constant coefficient, Solution of difference equation-complementary function-Particular integral, Z - transform, Properties of Z-transforms, Convolution of two sequences, inverse Z-transform, Solution of Difference equations using Z-transform.

Unit III: Fourier Transform **Hours: 12**

Fourier Integral, Fourier Cosine and Sine transform, Discrete Fourier Transform, Continuous Fourier Transform, Linearity, time shifting, frequency shifting, and convolution properties, Applications in image processing, communication systems, and quantum mechanics.

Unit IV: Partial Differential Equations **Hours: 12**

Formation of Partial Differential Equations (PDEs), Solutions of PDEs of the first order, Cauchy problem for first-order PDEs, Lagrange's method, Charpit and Jacobi methods for solving first-order nonlinear PDEs. Classification of second order PDEs, one-dimensional heat equation, wave equation, and Laplace equation, Applications of PDE's to wave analysis.

Max <52Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the partial differential equations of first and second order in order to model or understand the Engineering applications
CO2 - Understand the techniques of Laplace transforms, Z-transforms, and Fourier Transform to solve critical mathematical problems.
CO3 - Apply the methods of Transform techniques to solve differential equations involving piecewise continuous and exponential function.
CO4 - Classify the partial differential equations of different orders and solve them using different techniques
CO5 - Appraise engineering or scientific problems that can be addressed using transform techniques
CO6 - Formulate and create various application problems in engineering using series and Transform techniques

TEXT / REFERENCE BOOKS

1. Jain, R. K., & Iyengar, S. R. K. (2002). *Advanced engineering mathematics* (3rd ed.). Narosa Publishing House.
2. Kreyszig, E. (2016). *Advanced engineering mathematics* (10th ed.). John Wiley & Sons.
3. O'Neil, P. V. (2017). *Advanced engineering mathematics* (8th ed.). Cengage Learning.
4. Rao, K. S. (2011). *Introduction to partial differential equations* (3rd ed.). PHI Learning.
5. Amaranath, T. (2003). *An elementary course in partial differential equations* (2nd ed.). Narosa Publishing House.
6. James, G. (2016). *Advanced modern engineering mathematics* (4th ed.). Pearson.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE201T					Introduction to Artificial Intelligence					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Understand the core principles of AI and its various subfields.
- Identify the potential of AI for solving complex problems in petroleum engineering.
- Explore how AI techniques can be applied for data analysis and building basic AI models.
- Develop basic skills in utilizing AI tools for data analysis and decision-making.

Unit I: Fundamentals of Artificial Intelligence **Hours: 10**

Introduction to AI, Machine Learning (ML) vs. Deep Learning (DL), Bridging Python Programming, **Supervised Learning:** Algorithms (linear regression, decision trees), **Unsupervised Learning:** Techniques (clustering, dimensionality reduction), Reinforcement Learning, **Data Preprocessing:** Techniques for cleaning, transforming, and preparing data for AI models.

Unit II: AI Model Evaluation **Hours: 10**

Accuracy, Precision, Recall, F1-Score, Confusion Matrix, Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R-Squared (Coefficient of Determination), Area Under the ROC Curve (AUC-ROC). **Examples:** water-cut prediction accuracy, Net Present Value (NPV) improvement from reservoir management strategies, or drilling cost reduction achieved through AI-powered optimization.

Unit III: Introduction to Artificial Neural Networks (Ann) **Hours: 10**

Biological Inspiration, Artificial Neurons, Network Architectures, Activation Functions, Learning Processes, Loss Functions, Overfitting and Underfitting, Deep Learning Architectures, and Optimization Algorithms. **ANN applications in Petroleum Engineering:** Reservoir property prediction using CNNs, production forecasting with RNNs, or wellbore stability analysis with MLPs.

Unit IV: Time Series Analysis and Ethics **Hours: 09**

Artificial time series analysis, Application of time series prediction in oil and gas industry, Ethical considerations of AI, Explainability and Transparency, Environmental Impact, Data Privacy and Security, Long-term Sustainability, The Future of AI Ethics.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Define key terms and concepts related to Artificial Intelligence (AI), machine learning (ML), and deep learning (DL)

CO2- Understand the fundamental differences between supervised, unsupervised, and reinforcement learning techniques, and identify their potential applications

CO3- Apply basic machine learning algorithms (e.g., linear regression, decision trees)

CO4- Evaluate the performance of different AI models

CO5- Assess the limitations and ethical considerations associated with utilizing AI models in the oil and gas industry

CO6- Design a simple AI workflow for a specific petroleum engineering problem

TEXT / REFERENCE BOOKS

1. Hemmati-Sarapardeh, A., Husein, M. M., Karan, K., & others. (2020). *Applications of artificial intelligence techniques in the petroleum industry*. Gulf Professional Publishing.
2. Pandey, Y. N., Rastogi, A., Kainkaryam, S., Bhattacharya, S., & Saputelli, L. (2020). *Machine learning in the oil and gas industry: Including geosciences, reservoir engineering, and production engineering with Python*. Apress.
3. Vishwas, B., & Patel, A. (2020). *Hands-on time series analysis with Python: From basics to bleeding edge techniques*. Apress.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

PART A: 10 Questions of 2 marks each-No choice **20 Marks**

PART B: Each unit carries 20 marks, with internal choice **80 Marks**

25PE201T					Fundamentals of Fluid Mechanics for Petroleum Engineers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- Inculcate the importance of fluid statics and dynamics.
- Acquire a sound knowledge on fluid properties and its transport
- Familiar with various flow measurement and transportation devices.

Unit I: Fundamentals of Fluid Mechanics (Introduction and Fluid Statics) **Hours: 10**

Definition and Classification of Fluids-Real and ideal fluids, continuum hypothesis, viscosity, compressibility. Fluid Properties: Density, specific gravity, viscosity, surface tension, vapour pressure, and compressibility factor. Fluid Statics: Pressure and its measurement (manometers, Bourdon gauge, piezometer). Variation of pressure with depth. Hydrostatic forces on submerged plane and curved surfaces. Buoyancy, stability of floating bodies.

Unit II: Fluid Kinematics and Dynamics **Hours: 10**

Kinematics of Fluid Flow: Types of flow – steady/unsteady, uniform/non-uniform, laminar/turbulent, rotational/irrotational. Streamlines, Pathlines, and Streaklines. Continuity Equation. Euler's and Bernoulli's Equations: Derivation, assumptions, and practical applications. Energy and Momentum Principles: Momentum equation and applications (nozzle, diffuser, bend, and jet impingement).

Unit III: Fluid Flow Through Pipes and Porous Media **Hours: 10**

Flow in Pipes: Laminar and turbulent flow in circular pipes. Hagen–Poiseuille and Darcy–Weisbach equations. Friction factor correlations. Minor losses – fittings, bends, expansions, contractions. Flow Through Porous Media: Darcy's Law, permeability, and porosity concepts. Non-Darcy (Forchheimer) flow. Capillary pressure and relative permeability. Measurement of permeability in core samples.

Unit IV: Flow Measurement in Petroleum Systems **Hours: 09**

Principles and importance of flow measurement in petroleum operations; types of flow meters and measuring devices – Venturimeter, Orifice meter, Rotameter, Turbine and Positive Displacement meters – construction, working principles, calibration, and applications. **Valves** – types (gate, globe, ball, check, butterfly), functions, and flow control characteristics. **Pumps and compressors** – classification, working principles, performance characteristics, and selection criteria for petroleum.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the properties of fluids and their influence on fluid motion
CO2 - Illustrate the kinematics and dynamics of fluid flow.
CO3 - Apply Euler's and Bernoulli's equation in various flow systems.
CO4 - Estimate the energy losses in pipeline system.
CO5 - Understand the design of various flow measurement instruments
CO6 - Evaluate the performance characteristics of pumps and compressors

TEXT / REFERENCE BOOKS

1. Bansal, R. K. (2024). *A textbook of fluid mechanics and hydraulic machines* (11th ed.). Laxmi Publications.
2. De Nevers, N. (2021). *Fluid mechanics for chemical engineers* (4th ed.). McGraw-Hill.
3. Munson, B. R., Okiishi, T. H., Huebsch, W. W., & Rothmayer, A. P. (2020). *Fundamentals of fluid mechanics* (9th ed.). John Wiley & Sons.
4. McCabe, W. L., Smith, J. C., & Harriott, P. (2012). *Unit operations of chemical engineering* (7th ed.). McGraw-Hill Education.
5. White, F. M. (2022). *Fluid mechanics* (9th ed.). McGraw-Hill Education.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE203T					Thermodynamics of Petroleum Reservoir Fluids					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	1	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To provide the understanding of fundamentals of thermodynamics of reservoir fluids and their phase behaviour.
- To provide the concept and understanding of fluid sampling and PVT study.
- To provide the understanding of compositional characterization and application of various correlations in real field and simulation application.

Unit I: Reservoir Fluids and Hydrocarbon Phase Behaviour **Hours: 10**

Reservoir and reservoir fluids, Hydrocarbon-formation in source rock and crude oil in reservoirs, Thermodynamic behaviour –single, two, three and multicomponent system. Physical properties of petroleum reservoir fluids (Oil and water), classification of reservoirs and reservoir fluids, Compositional analysis of crude oil, Assignments and tutorials.

Unit II: Properties Hydrocarbon Components, Characterization and Correlation **Hours: 10**

Natural gas properties, behavior of ideal and real gases. Characterizing Hydrocarbon-plus fractions: generalized correlations, PNA determination, splitting and lumping scheme including various correlation methods, Assignments and tutorials

Unit III: Sampling, PVT Properties and Laboratory Study of PVT **Hours: 10**

Collection of reservoir fluid samples for PVT study, PVT analysis: Constant composition expansion, flash liberation, differential liberation, separator test for PVT data of hydrocarbon fluids. Evaluation and correlation of physical and chemical properties of reservoir fluids including laboratory and empirical methods, Assignments and tutorials.

Unit IV: Selection of PVT Correlations **Hours: 09**

Vapour-liquid equilibrium calculation, Selection of PVT oil properties correlations, Correlation selection based on geographical location, type of fluid, PVT model, data range and ranking and expert system or set of rules, Applications for PVT correlation selection.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the compositional range of hydrocarbon components present in reservoir fluids with crude typing.
CO2 - Understand hydrocarbon phase behaviour in dynamic reservoir conditions.
CO3 - Apply knowledge for chemical characterization of hydrocarbon reservoir fluids and available correlations.
CO4 - Design sample collection for various purposed and understand PVT study and analysis.
CO5 - Analyse situation dependent applicability of different correlation of vapor liquid equilibrium calculation.
CO6 - Apply the knowledge for selection of PVT correlations.

TEXT / REFERENCE BOOKS

1. Ahmed, T. (2016). *Equation of state and PVT analysis: Applications for improved reservoir modeling*. Gulf Professional Publishing.
2. Firoozabadi, A., & Venkatesan, R. (2018). *Thermodynamics of hydrocarbon reservoirs*. McGraw-Hill Education
3. Danesh, A. (1998). *PVT and phase behaviour of petroleum reservoir fluids*. Elsevier.
4. Dandekar, A. Y. (2013). *Petroleum reservoir rock and fluid properties* (2nd ed.). CRC Press.
5. El-Banbi, A., Alzahabi, A., & El-Maraghi, A. (2018). *PVT property correlations: Selection and estimation*. Gulf Professional Publishing.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE204T					Geomechanics and Strength of Materials					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	1	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To learn stress and strain concept with numerical problems and mechanical properties of material and learn the concept of material failure
- To study strength of geological material such as rock and soil filled with or without pore fluid and applicable failure theories
- To understand the tectonic stress field and stress classification in subsurface
- To understand the effect of pore pressure at depth and various laws governing strength of rock in the presence or absence of pore fluid

Unit I: Basics of Mechanics **Hours: 10**

Fundamental assumptions in elementary mechanics of materials; viscoelasticity or rheology; mechanical properties of metals – concepts of stress and strain; stress-strain behavior; inelasticity; elastic properties of materials; tensile properties; elastic recovery during plastic deformation; compressive, shear, and torsional deformation; hardness; variability of material properties; design/safety factors. Testing of material properties. Thermal stresses – bars subjected to tension, asymmetric loading, and stress calculation of cylindrical bodies. Scalar, vector, tensor; composition and resolution of stresses; stress vector acting on a surface; analysis of stress; stress field acting at a point – derivation of equations for σ_n and τ_n. Mohr's circle; deviatoric and mean stress; special states of stresses by Mohr's circle; Mohr-Coulomb failure envelope.

Unit II: Introduction to Rock Mechanics **Hours: 10**

Structural rock mechanics; strength of geological materials; influence of geological history (burial and uplift) on rocks – normally consolidated and over consolidated rocks; effective stress, undrained modulus of elasticity, coefficient of compressibility, coefficient of consolidation, significance of drainage path length and the rate of consolidation; stress and strain in geological materials and their measurements; cohesion, friction, and failure; influence of fabric; drained and undrained strength; Atterberg limits

Unit III: Introduction to Reservoir Geomechanics **Hours: 10**

Tectonic stress field; pore pressure at depth in sedimentary basins; basic constitutive laws; rock failure in compression; tension and shear; faults and fractures at depth; compressive and tensile failures in vertical wells; determination of S_3 from mini-fracs and extended leak-off tests and constraining the magnitude of S_H from wellbore failures in vertical wells; wellbore failure and stress determination in deviated wells; stress fields – from tectonic plates to reservoirs around the world; wellbore stability; critically stressed faults and fluid flow; effects of reservoir depletion

Unit IV: Application of Geomechanics **Hours: 09**

Hydraulic Fracturing: Process, geomechanical factors effecting fracture development and simple calculations; Application of Geomechanics in Subsurface Energy – Oil and Gas, Geothermal Energy, Carbon Capture and Storage, and Hydrogen Storage.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the structure and mechanical properties of different materials viz., metals, soil, minerals and rocks
CO2 - Determine the geomechanical properties of materials and predicting the failure based on the various failure theories
CO3 - Estimate the changes in shear strength of reservoir rocks vis-à-vis reservoir production, and its effect on the petrophysical properties
CO4 - Apply the understanding of stress-field to optimize production
CO5 - Evaluate the failure of the wall of the wellbore during drilling activities
CO6 - Analyze stress-field around boreholes using laboratory data and extending it to field situations.

TEXT / REFERENCE BOOKS

1. Zoback, M. D. (2010). *Reservoir geomechanics*. Cambridge University Press.
2. Longuemare, P. (2002). *Geomechanics in reservoir simulation*. Editions Technip.
3. Nauroy, J. F. (2011). *Geomechanics applied to petroleum engineering*. Editions Technip.
4. Bansal, R. K. (2023). *A textbook of strength of materials* (6th ed.). Laxmi Publications.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE204P					Geomechanics and Strength of Materials Practical					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To understand geomechanical properties of the reservoir/seal rocks
- Differentiate between strain energy and strain energy density
- To understand failure planes in different rocks

LIST OF EXPERIMENTS

1. To perform the hardness test on the test specimen by Rockwell and Brinell methods
2. To perform the impact test on the test specimen by Izod and Charpy methods
3. To perform the tension test on the test specimen
4. To perform the torsion test on the test specimen
5. To perform the bending test on the test specimen
6. To perform the shear test on the test specimen
7. To perform the compression test on the test specimen
8. To perform the fatigue test on the test specimen
9. To understand the triaxial test of reservoir rocks on the test specimen

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the structure and mechanical properties of different materials viz., metals, soil, minerals and rocks
CO2 - Determine the geomechanical properties of materials and predicting the failure based on the various failure theories
CO3 - Estimate the changes in shear strength of reservoir rocks vis-à-vis reservoir production, and its effect on the petrophysical properties
CO4 - Apply the understanding of stress-field to optimize production
CO5 - Evaluate the failure of the wall of the wellbore during drilling activities
CO6 - Analyze stress-field around boreholes using laboratory data and extending it to field situations

TEXT / REFERENCE BOOKS

1. Zoback, M. D. (2010). *Reservoir geomechanics*. Cambridge University Press.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24PE205T					Petroleum Exploration					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To provide the fundamental of petroleum exploration, different methods of geophysical surveys and instruments used for it.
- To provide the understanding of geophysical data interpretation and analysis
- To provide how to examine the acquired data to understand the position and extent of subsurface prospects in terms of depth and aeri

Unit I: Fundamentals of Petroleum Exploration **Hours: 10**

Ingredients of Petroleum Exploration, Concept of source, reservoir, migration, trap and seal, Concept of Play, Lead, Prospect and Drillable Prospect, Types of Petroleum Traps-Structural, Stratigraphic and Combinational traps, Primary and Secondary Migration, E&P Life Cycle, Concept of Reserve, Lease and Reservoir, Techniques of Petroleum Exploration, Geochemical, Gravity, Magnetic, Electrical and seismic method of hydrocarbon exploration, Palaeontology & Palynology in Exploration

Unit II: Geochemical Analysis **Hours: 10**

Petroleum geochemistry in exploration, Burial history curve with oil and gas window. Seeps and surface geochemical exploration, Rock eval pyrolysis and maturity assessment, Fluid inclusion studies, Radiometric age dating, Geochemical welllogs for exploration, Molecular approach for petroleum exploration.

Unit III: G and M Methods **Hours: 09**

Gravity and magnetic prospecting, Instruments of G&M survey, Gravity and magnetic data correction, Interpretation of G&M anomaly, Correlation of Gravity anomaly with seismic anomaly. SP, Telluric and Magnetotelluric data interpretation, Electrical properties of hydrocarbon, Electrical conductivities, Resistivity's of various lithologies, Dielectric constants, Land airborne EM, Interpretation and Modeling of data, Potential estimation for various buried bodies, Anomaly and well placement based on electrical data. Basic well logs, GR

Unit IV: Fundamentals of Seismic Processing, Interpretation and Attribute **Hours: 10**

Body waves and surface waves, Rayleigh, Love, P and S wave, Seismic acquisition principle, Seismic refraction and reflection surveys, Land and marine sources, Geophone, Hydrophone and Vibrose is survey, Seismic Fold, Signal and Noise, Seismic Processing, SEG D and SEG Y format, CDP/CMP and NMO, DMO, Seismic migration, Base map, Strike Line and Dip Line, 2D and 3D seismic, inline and cross line, 3D fold, time slice and its importance. Horizon and Fault mapping, Seismic impedance and reflection coefficient, convolution and autocorrelation, Fault skeleton preparation, wrench system Structural and Stratigraphic interpretation, Synthetic generation, Time and depth map, VSP survey, Attributes: Amplitude, Frequency and Sweetness, AVO analysis, Classification of sands, Rock solid attributes, Hydrocarbon Resource Reserve Estimation.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Evaluate prospects and drillable prospects.

CO2 - Understand the petroleum system along with the source analysis

CO3 - Examine the Process of gravity, magnetic, seismic and resistivity data acquisition, processing and interpretation will help students to handle instruments like gravimeter, magnetometer, seismometer, seismometer and resistivity meter

CO4 - Integrate gravity and magnetic data to understand the density and magnetic susceptibility of the subsurface.

CO5 - Integrate seismic, well log and other geophysical data for volume estimation.

CO6 - Examine the structural and stratigraphic data to understand the position and extent of subsurface prospects in terms of depth and aerial extent. Determine in stochastic sense hydrocarbon resource reserve estimation.

TEXT / REFERENCE BOOKS

1. Sengupta, S. M. (2017). *Introduction to sedimentology* (3rd ed.). CRC Press/Balkema.
2. Gadallah, M. R. (1994). *Reservoir seismology*. PennWell Books.
3. Telford, W. M., Geldart, L. P., Sheriff, R. E., & Keys, D. A. (1990). *Applied geophysics* (2nd ed.). Cambridge University Press.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

PART A: 10 Questions of 2 marks each-No choice **20 Marks**

PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE205P					Petroleum Exploration Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To provide the fundamental of petroleum exploration, different methods of geophysical surveys and instruments used for it.
- To provide an understanding of geophysical data interpretation and analysis.

LIST OF EXPERIMENTS

1. Basics of Contouring (Hand Drawn and Computerised)
2. Seismic Data Acquistion, Processing
3. Seismic Data Interpretation
4. Horizon and Fault Mapping of processed 2D line
5. Identification of Prospects and Drillable Prospects on a Seismic Section
6. Gravity data Acquisition
7. Gravity Data Processing and Interpretation
8. Magnetic Data Acquisition
9. Magnetic Processing and Interpretation
10. Resistivity Data Acquistion
11. Resistivity Data Processing and Interpretation

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basics of contouring

CO2 - Understand the petroleum system along with the source analysis

CO3 - Examine the Process of gravity, magnetic, seismic and resistivity data acquisition, processing and interpretation will help students to handle instruments like gravimeter, magnetometer, seismometer, seismometer and resistivity meter

CO4 - Integrate gravity and magnetic data to understand the density and magnetic susceptibility of the subsurface.

CO5 - Integrate seismic, well log and other geophysical data for volume estimation

CO6 - Evaluate prospects and drillable prospects.

TEXT / REFERENCE BOOKS

1. Gadallah, M. R. (1994). *Reservoir seismology*. PennWell Books.
2. Telford, W. M., Geldart, L. P., Sheriff, R. E., & Keys, D. A. (1990). *Applied geophysics* (2nd ed.). Cambridge University Press.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**